

SHORT VS. LONG INTERVALS IN PRO CYCLISTS

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In the last [post](#) we saw that short intervals (3x13min 30s on-15s off) provides superior physiological and performance improvements compared to long intervals (4x5min on with 2.5 recovery). That study was performed on **well-trained** (VO2max ~66) but not professional cyclists.

Would the same results be obtained with **professional athletes**?

Are short intervals superior to long intervals even for pro cyclists?

A study published by the same research group lead by Prof. Bent Rønnestad on *Scandinavian Journal of Medicine & Science in Sports* in 2020 tried to answer this question.

WHAT DID THEY DO?

The research design was the same of the research published in the last post, the only differences were the following:

1. Participants were pro and not well-trained cyclists

2. Training intervention lasted 3 and not 10 weeks.

3. High intensity training was performed 3 and not 2 times per week.

So, here what they did:

- 18 professional cyclists (on average ~25 years old and ~73 VO₂max) were divided into two groups: SI (Short Intervals) and LI (Long Intervals).

- For the following 3 weeks the two groups performed the same training program:

8-10 hrs of low intensity training + 3x week HIT sessions.

The only difference between the two groups was about the HIT format they performed:

- **SI group performed Short intervals HIT sessions:**

3x13' alternating 30" on/15" off with 3' recovery in between.

- **LI group a Long Intervals HIT sessions:**

4x5' on/2.5' off.

15-20min of warm-up and cool down were included in each session.

If you make some calculations, you can see as both SI and LI groups accumulated a similar time in the 'on' phase: 19min30s vs 20 min (3x13x30" vs 4x5').

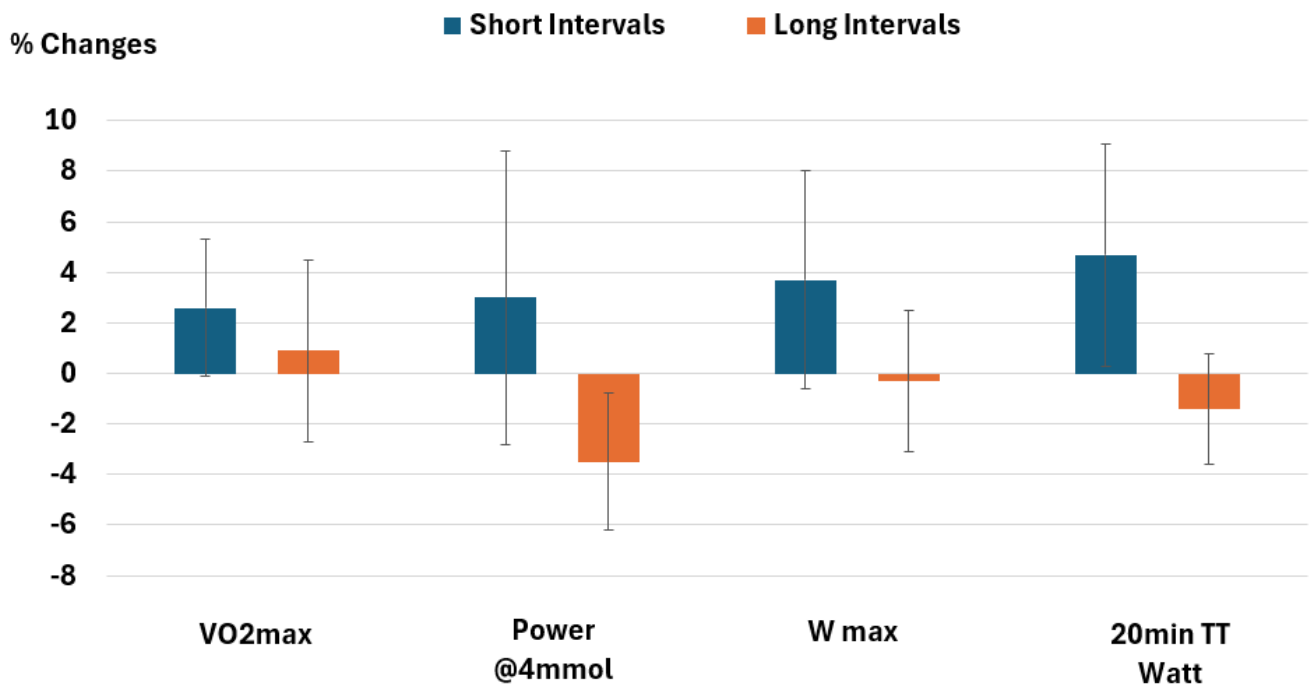
In both HIT's formats, the 'on' phases were performed at the maximum sustainable effort to obtain the highest average power throughout all the intervals.

The 'off' phases were performed at 50% of the power output sustained during intervals, so it was active recovery (read high zone 1 or low zone 2).

WHAT DID THEY FOUND?

Globally, after the 3 training weeks. the SI group improved more the physiological and performance parameters compared to the LI group:

- VO₂max: 2.6±2.7% vs 0.9±3.6%;
- Power output at 4mMol/L: +3.0±5.8 vs -3.5±2.7 %
- Wmax (power output in the last step of a maximal incremental test):
3.7±4.3 vs -0.3±2.8%
- 20-min time trial: 4.7±4.4 vs -1.4±2.2%



% changes pre-post the 3 training weeks in physiological and performance parameters in the short intervals vs long intervals groups. Data from Rønnestad et al (1)

The decrease in power @4mmol and 20min TT Watt in the long intervals group can be related to the decreased training volume compared to before starting the three weeks intervention: 10hrs per week during the 3 weeks intervention vs 18hrs per week before the 3 weeks intervention. However, the same volume decrease was performed by the short intervals group's cyclists, so this wasn't a factor confounding the results when interpreting the differences in performance improvements between the two groups.

Note that, as a result of performing the intervals at the maximum sustainable power output, **the SI group average power during the 'on' phases was higher compared to the LI group: 441 W (~130% second threshold) vs 368 W (~115% second threshold).** Despite this, a similar rating of perceived exertion (RPE) between groups, both when averaged across all the intervals and in the last interval, was reported.

PRACTICAL TAKEAWAYS

In professional cyclists, 3x13min alternating 30s on and 15s off seems to provide superior physiological and performance adaptations compared to 4x5min intervals with 2.5min recovery, when both formats are performed at the maximum sustainable intensity.

GABRIELE'S COMMENT

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- While in the previous post, we saw that short intervals seems to be a better solution to improve performance when compared to long intervals in trained cyclists, **this study suggests that this could be also extended to professional cyclists**. Again, we are not sure about the physiological mechanism determining the adaptations' differences. A hypothesis could be that, **when matching the perceived exertion, short intervals permit to accumulate more time at higher power output (and so maybe to stimulate a stronger biological signal for physiological adaptations) when compared to long intervals**. However, this remains just a speculation with no data to support it.
 - The widespread training principles of individualization, progressive overload and specificity should be considered also when applying these data on professional cyclists:
 - **Individualization**: as every athlete is unique, I would suggest trying both HIT formats for each individual and then looking at which one he/she can sustain a higher power output and heart rate for a given total duration and perceived effort. This is probably the HIT most effective for him/her. Anyway, it is very likely that it will be the short intervals' case.
 - **Progressive overload**: when starting to put short intervals in the training plan, it could be reasonable to start with a lower number of repetition than the 3x13min used in this study (e.g. 3x8'), and then progressively increase the number of repetitions workout after workout to arrive smoothly at the 3x13' used in this study. You could also start with a 1:1 work:recovery ratio instead of 2:1 (so alternating 30s 'on' and 30s 'off', instead of 30s 'on' and 15s 'off').
 - **Specificity**: even if short intervals seem generally better to obtain adaptations both for short and long duration performance, it could make sense to use long intervals in the last weeks before the goal race, when they represent an effort similar to what will be performed during the competition. For example the track individual pursuit's effort is very similar to long intervals.

Thanks for Reading! Spread with your Team!

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REFERENCES:

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1. Rønnestad BR, Hansen J, Nygaard H, Lundby C. Superior performance improvements in elite cyclists following short-interval vs effort-matched long-interval training. Scand J Med Sci Sports. 2020 May;30(5):849-857.





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